STAFF REVIEW · H446 224 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It marks the end of the list.

2. Into the left subtree.

3. Create and return a new node at that null position.

4. A right-leaning chain 10→20→30.

I Do — source worked steps

1. Predict first — Compare 12 with 20, then 10.

2. Model the reasoning — Insert 12 as the right child of 10, preserving the BST property.

We Do — guided CHECK

1. Model the reasoning
35<40 leads left, placing 35 as the left child of 40.

You Do — source mark schemes

1. [4] 12<15 [1], moves left to 9 [1], 12>9 [1], so it is right child of 9 [1].

2. [6] Parent of 35 is 40 [1], it is left child [1]; in-order sequence 20,30,35,40,50,70 [4].

3. [6] Find node 10 by moving left twice [2]; it is a leaf [1], so set parent 20.Left=null [1]. For a node with two children, replace it using an in-order successor/predecessor and remove the replacement from its old location [2].

Exit Ticket — source answer

1. True. Left subtree, node and right subtree produce increasing keys.